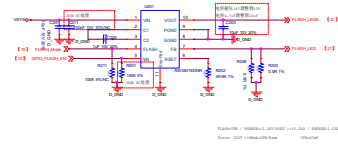
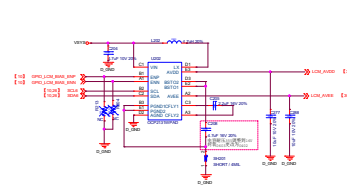


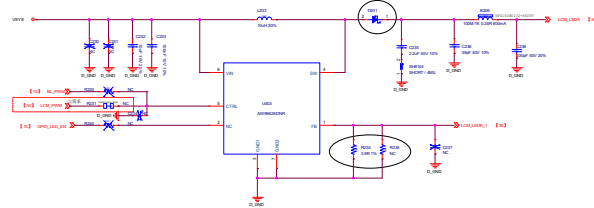
FLASH LED Driver



LCM BIAS



LCM LED Driver

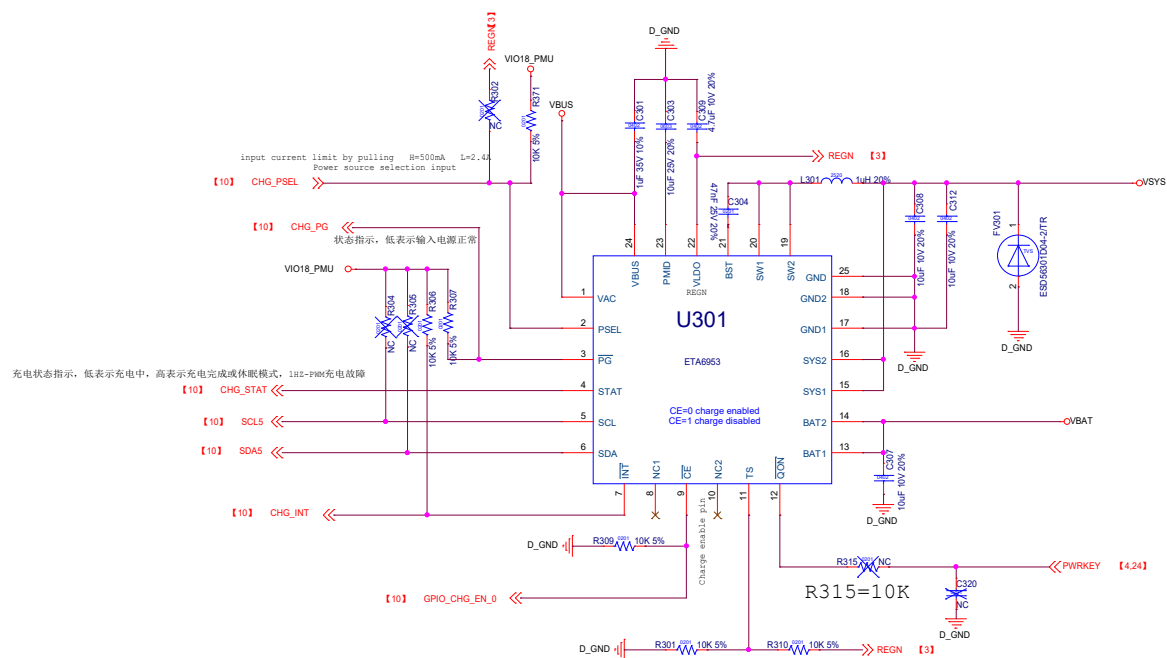


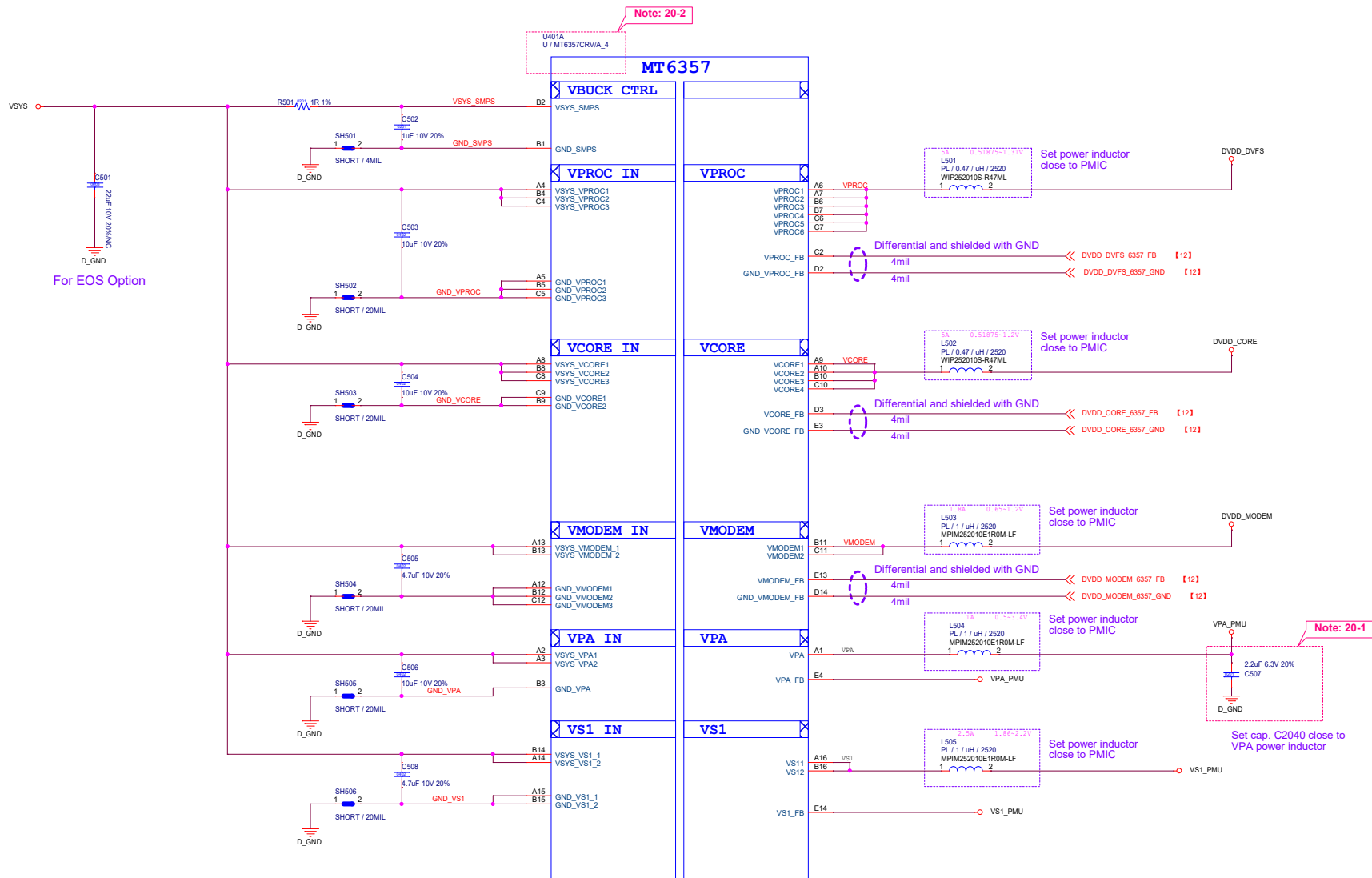
Switching Charger Power Path

ETA6953:address: D6H BQ25601:address: 6BH

FG

Open drain active low power good indicator. Connect to the pull up rail through 10-k Ω resistor. LOW indicates a good input source if the input voltage is between UVLO and ACOV, above SLEEP mode threshold, and current limit is above 30 mA.





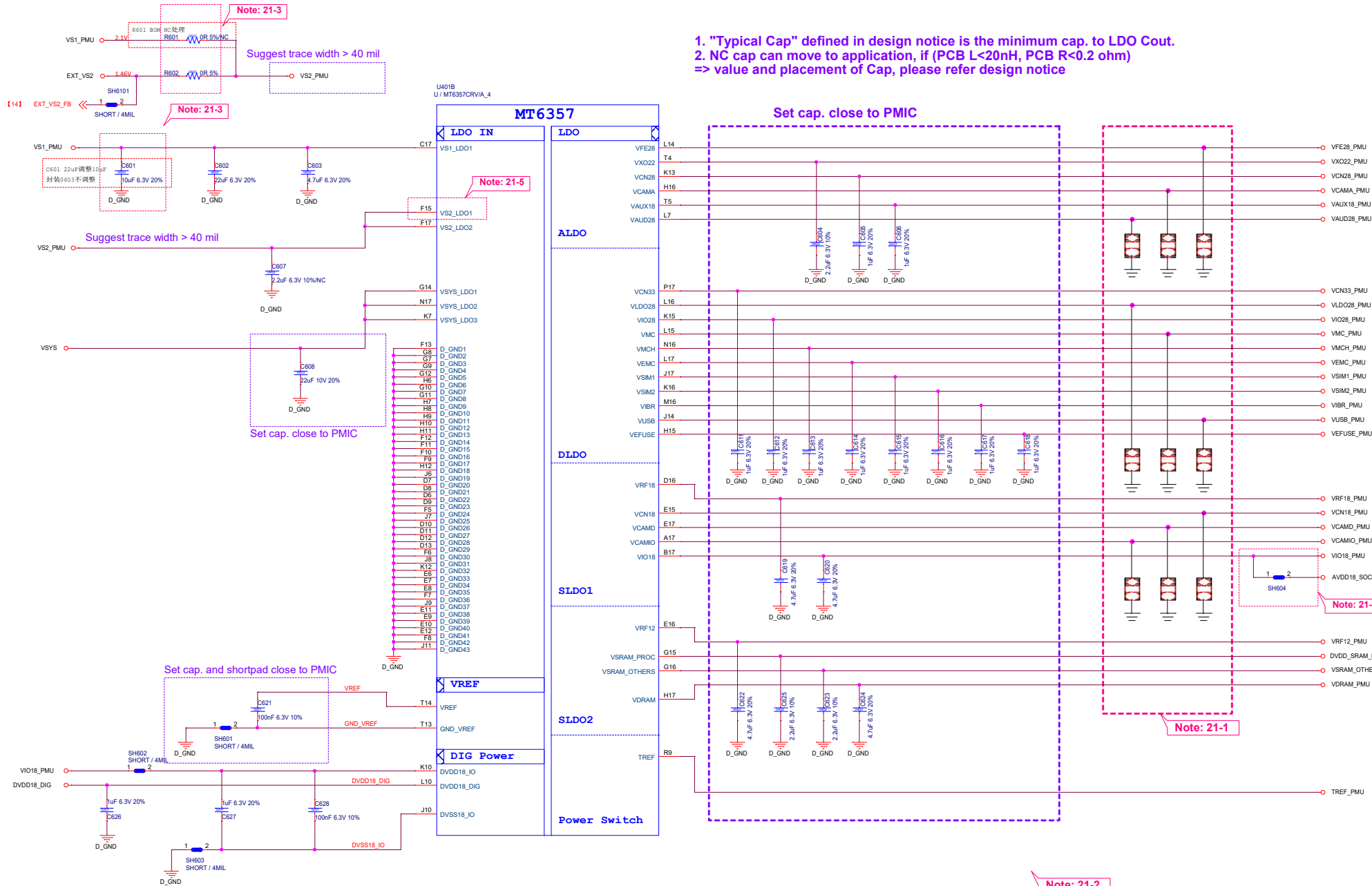
Schematic design notice of "20_POWER_MT6357_Buck"

Note 20-1: C2040, please choose 0402 size

Note 20-2: PMIC Part number notice for MT6765/62/61 platform

MTK Platform	PMIC
MT6765 / 62	MT6357 CRV
MT6761	MT6357 MRV

MEDIATEK	
Title	05_POWER_MT6357_Buck
Size	MTK Confidential
Date	Friday, August 26, 2022
Sheet	5 of 37



Schematic design notice of "21_POWER_MT6357_LDO"

Note 21-1: If these power trace can meet LDO layout constraint, these CAP can be NC or removed. Please refer to MT6357 design notice.

Note 21-2: Output cap range please follow MT6357CRV LDO design notice

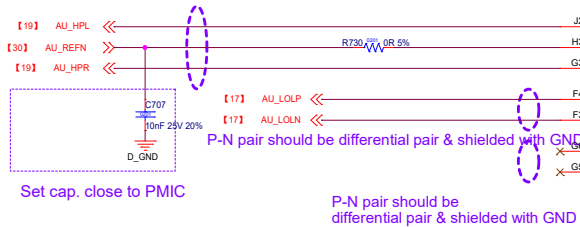
Note 21-3: Ext Buck BOM option

	Ext. buck option	
	w/ EXT VS2 Buck	N- EXT VS2 Buck
C601	10uF(0402)	22uF(0603)
R602	0-ohm , 0603	NC
R601	NC	0-ohm , 0603

Note 21-4: Please set SH2101 close to C2141, making star connection between VIO18_PMU and AVDD18_SOC near to LDO cap. C2141
Please also refer to MT6357 design notice for further detail design information

Note 21-5: Please connect VS2_LDO1(F15) to VS1_PMU if voltage applied to VCAMD(E17) > 1.3 V

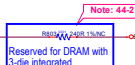
- AU_HPL and AU_HPR should be routed as single end signal, and be guarded by GND, up and down, left and right respectively
 - The suggested layout pattern of AU_HPL/ AU_HPR/ AU_REFN is " GND AU_HPL AU_REFN AU_HPR GND"

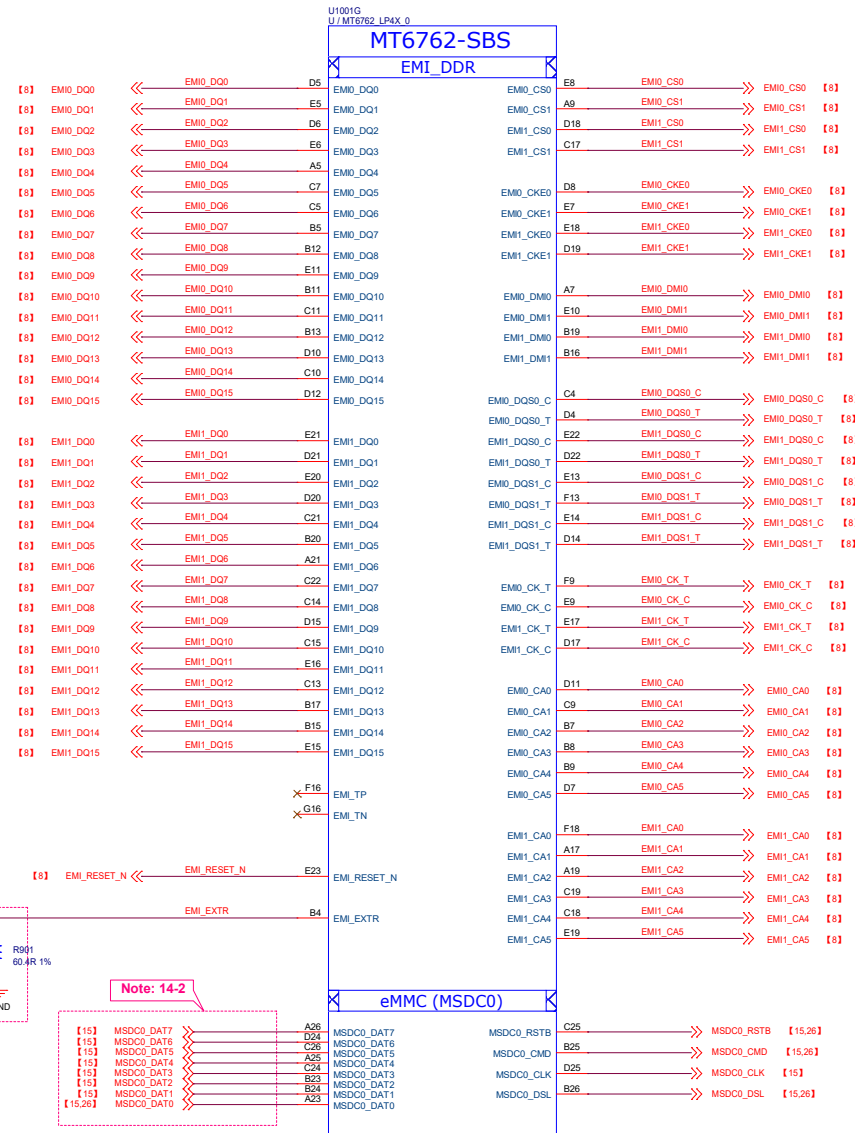


Set cap. and shortpad close to PMIC

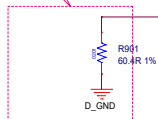
Set cap. close to PMIC

1. AVSS18_AUD is connected to GND in very short trace
2. AVSS18_AUD is connected to de-couple cap of AVDD18_AUD and AU_V18N with 6mil trace respectively

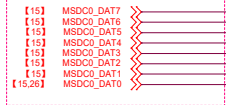




Note: 14-1



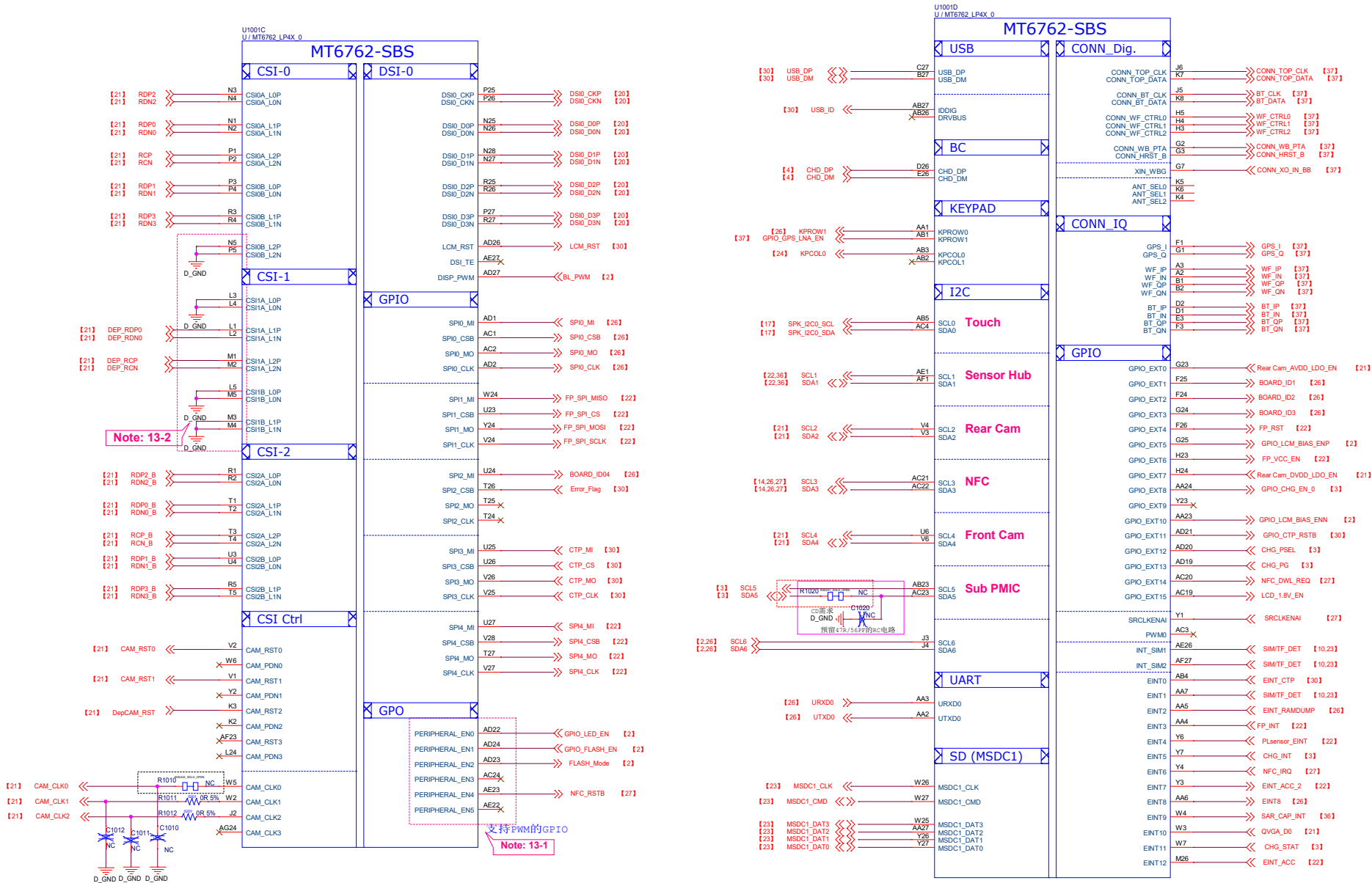
Note: 14-2



Schematic design notice of "14_BB_3" page.

Note 14-1: R4001 please select 60.4 ohm (1%) resistor

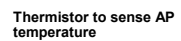
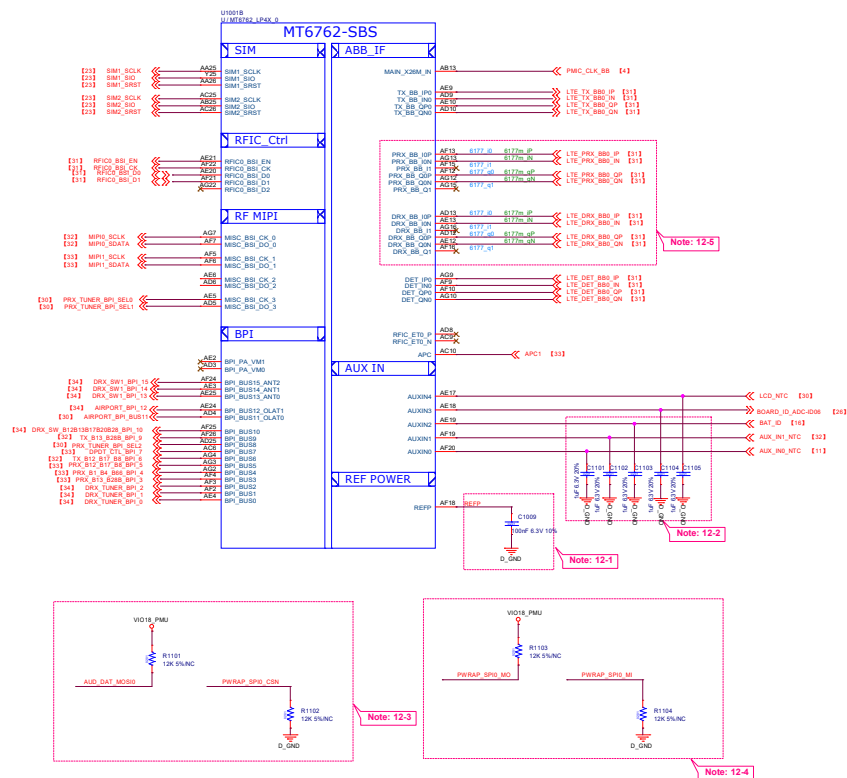
Note 14-2: Please check eMCP LP3 and eMCP LP4X/LP4 pin mux



Schematic design notice of "13_BB_2" page.

Note 13-1: The enable pin of acoustic or optoelectronic devices (e.g. SPK AMP/Backlight/Charger OCP/OVP) suggest to use Peripheral_EN[0:5]
If use other GPIOs as enable pin, suggest to reserve 0201 NC to GND

Note 13-2: MIPI CSI unused pin can be floating, but need correct SW setting



1. NTC1501 must keep a distance about 6-8 mm away from AP and far from other heat sources 10 mm at least.
2. The distance is the shortest distance from rack edge to edge.

Note 12-1: The de-coupling cap. for REFP (AF18 ball) have to be placed as close to BB as possible.

Note 12-2: To shunt a 1uF capacitor in the AUXIN ADC input to prevent noise coupling. It should be placed as close to BB as possible. Connect the unused AUX ADC input to GND.

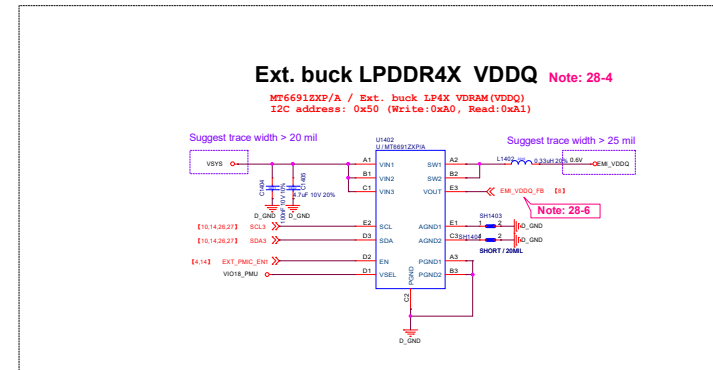
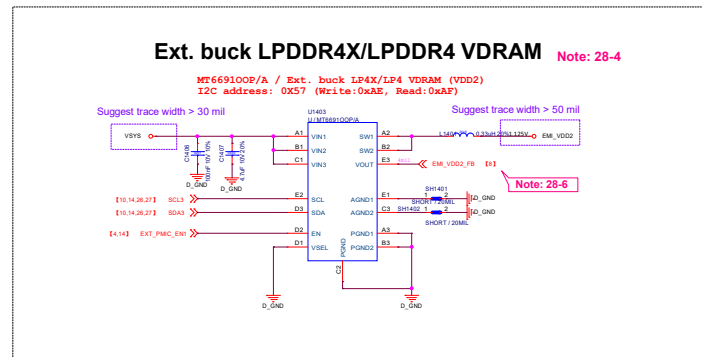
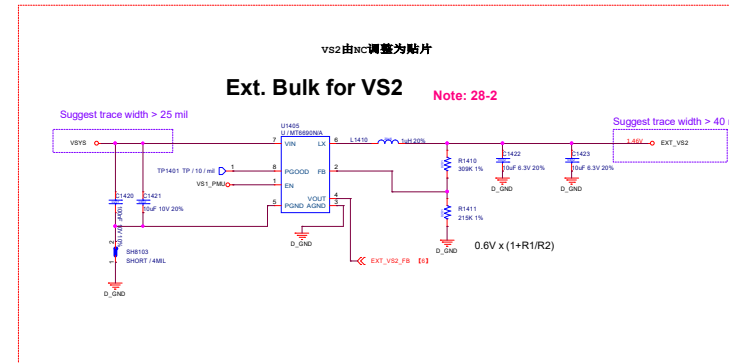
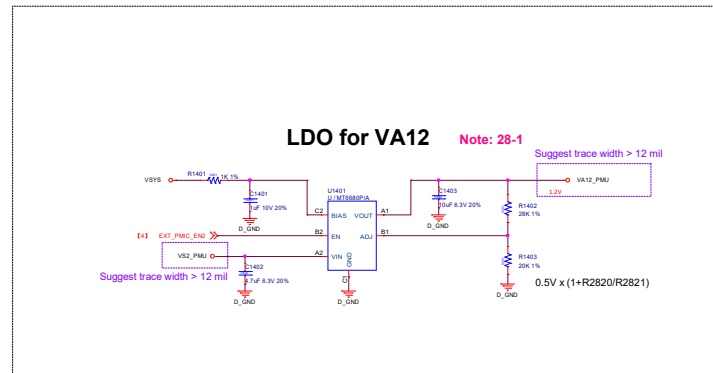
Note 12-3: "PWRAP_SPI0_CSN" and "AUD_DAT_MOSI0" are bootstrap pin to select which interface will be the JTAG pin out.

PWPRAP_SPI0_CSN		AUD_DAT_MOSIO		JTAG Function	
default=PU	default=PD	AP_JTAG	MD_JTAG		
HI	LO	N/A	N/A		
HI	HI (by ext. PU)	SPI0+EINT8	SPI1+SPI3		
LO (by ext. PD)	LO	SPI0+EINT8	N/A		
LO (by ext. PD)	HI	MSDC1	N/A		

Note 12-4: PWRAP_SPI0_MI / PWRAP_SPI0_MO is DDR type feature in bootstrap

PWRAP SPI0 MI	PWRAP SPI0 MO	Booting interface	
default=PU	default=PD	DDR	MSDC0 pin mux
LO (by ext. PU)	LO	LPDDR3	follow LPDDR3 Ref SCH.
LO (by ext. HI)	HI	N/A	N/A
HI (by ext. PU)	LO	LPDDR4X	follow LP4X/LP4 Ref SCH.
HI (by ext. HI)	HI	LPDDR4	follow LP4X/LP4 Ref SCH.

Note 12-5: Please set unused IQ pins in NC.



Schematic design notice of "28_POWER_ThirdParty-Power"

Note 28-1: VA12 Layout placement please close to AP

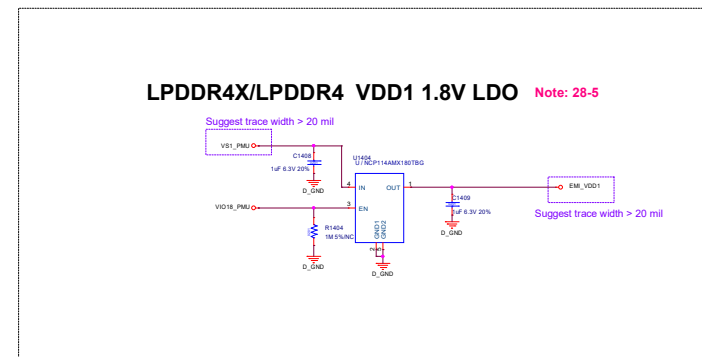
Note 28-2: VS2 Buck Layout placement please close to PMIC MT6357

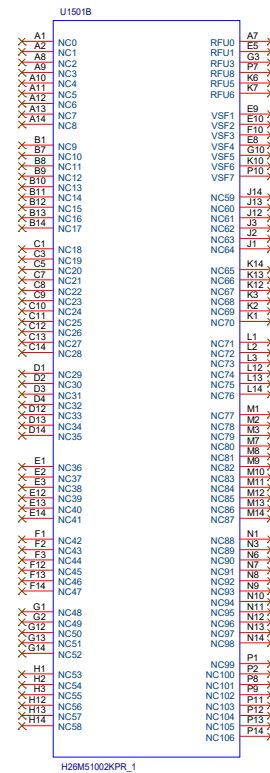
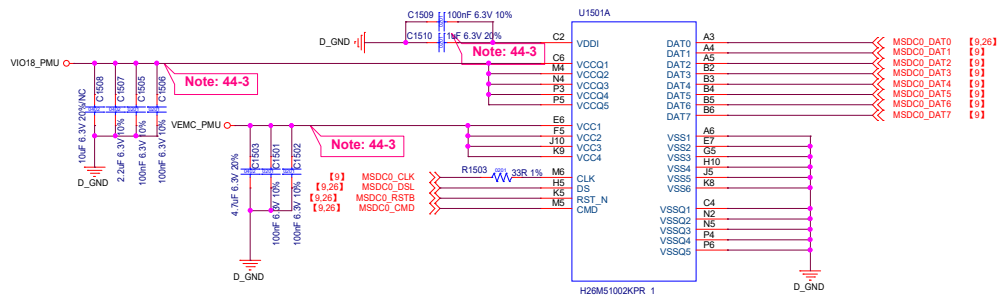
Note 28-3: VCN33 LDO Layout placement please close to MT6631

Note 28-4: 1.MT6691OOP/A Buck Layout Placement please close to LP4X/LP4
2.MT66912XP/A Buck Layout Placement please close to LP4X
3.If DRAM Application is LPDDR4 , MT66912XP/A NC

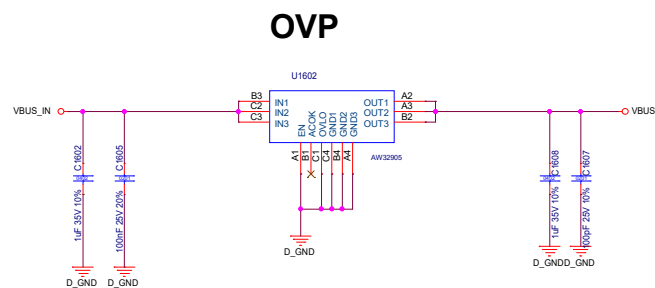
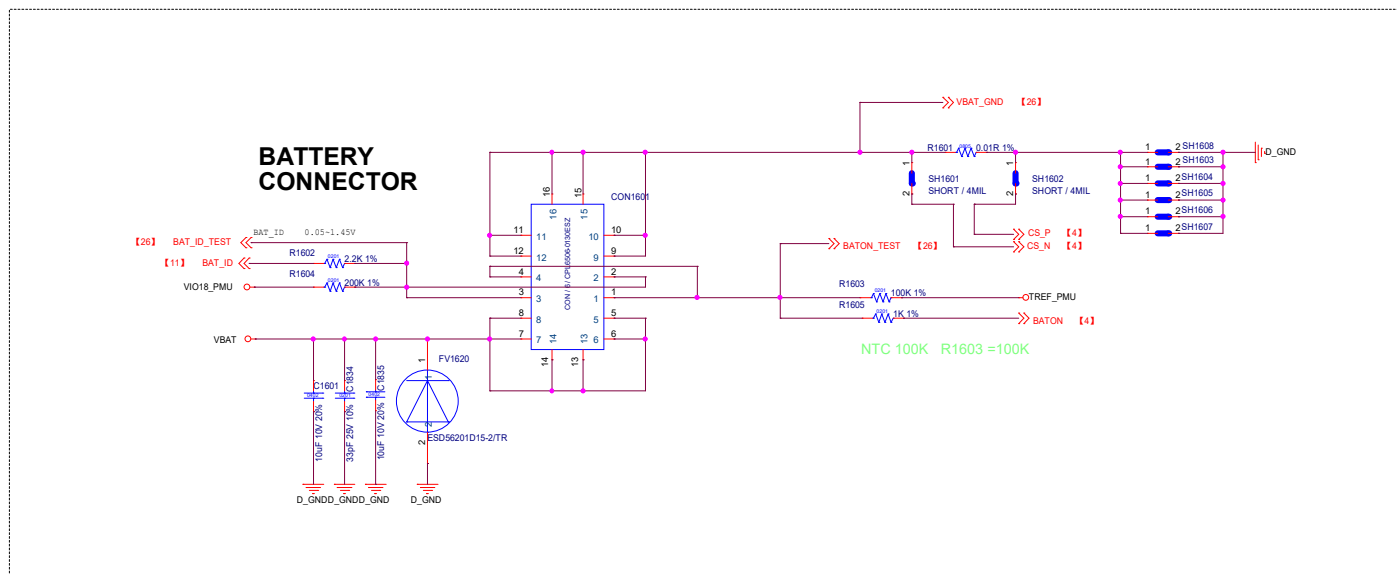
Note 28-5: U2810 LDO Layout Placement Please close to LPDDR4X/LPDDR4 VDD1 power ball

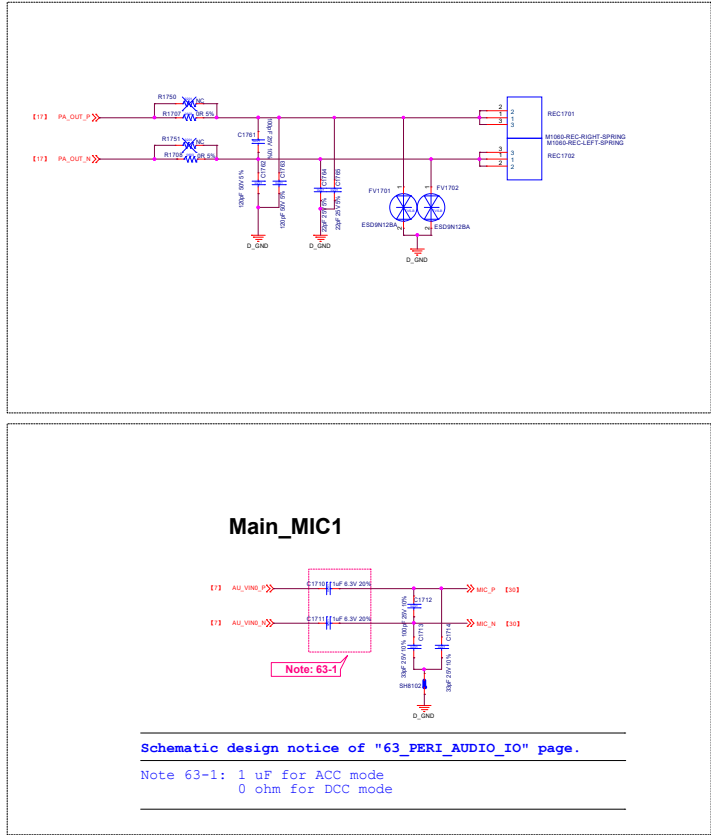
Note 28-6: For EM_VDD2_FB and EM_VDDQ_FB, please follow MMD rule



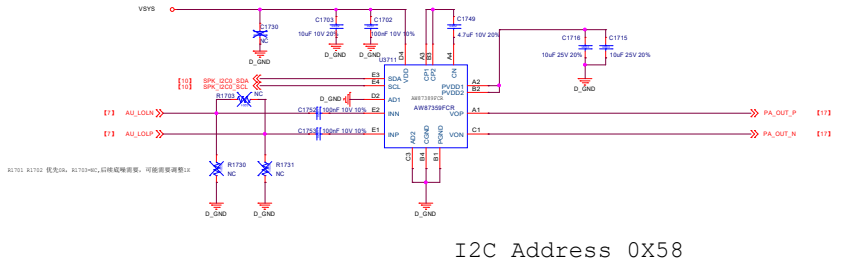


Note 44-3: Please refer to eMCP vendor's datasheet or MTK common design notice to get the recommendation bypass cap. value for VCC/VCCQ/VDDI power domains of eMMC.





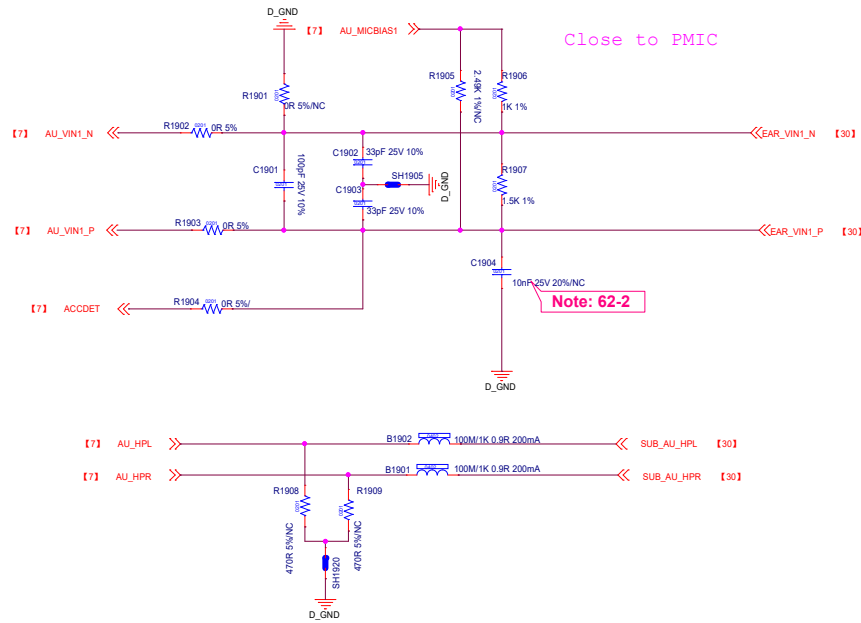
Audio PA



<Variant Name>

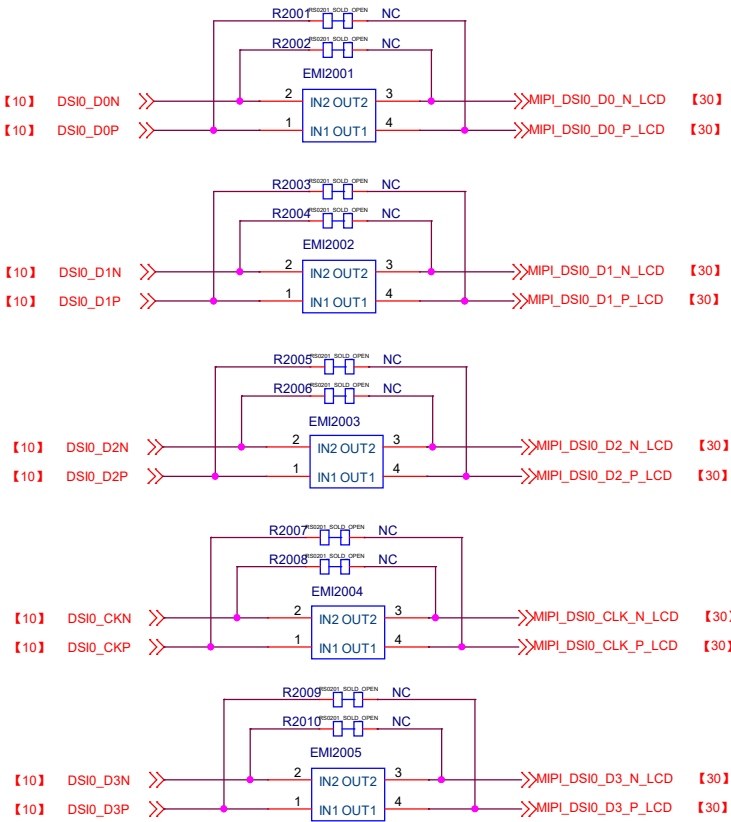
18.MIC Receiver	
MTK Confidential	
Date: 2022-08-28 10:37	

Earphone Audio



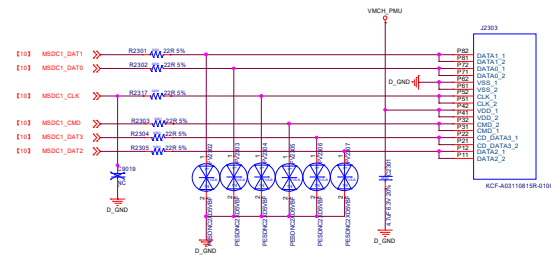
Note 62-2: Reserved Cap for CS/RS test, please double check multi-key function when used

Main LCM

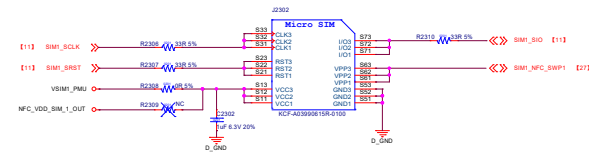


[illegible][illegible]

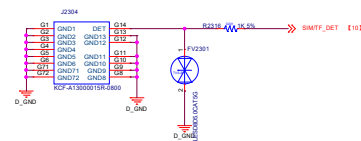
SD CARD



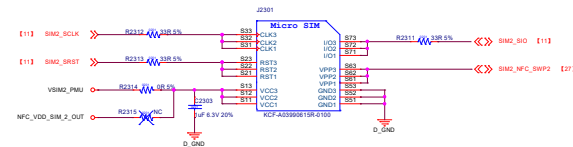
SIM1 PCB靠板边



Detect



SIM2 PCB靠板内



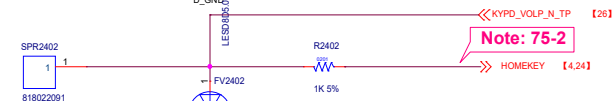
Power Key / Key Pad

DO NOT put pull-up resistor on PWRKEY

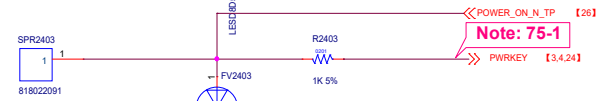
Volume Down



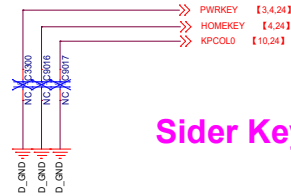
Volume Up



Power On



Sider Key



Schematic design notice of "65_PERI_Dual_SIM_ICUSB_KEYPAD" page.

Note 75-1: DO NOT put pull-up resistor on PWRKEY

Note 75-2: Volume Up : HOME Key / GND
Volume Down / FORCED_USB_BOOT : KPCOLD/ GND

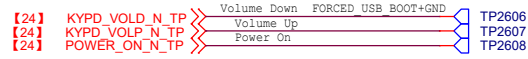
Note 75-3: For better ESD performance, please select suitable components for system protection.

	5	4	3	2	1
D					
C					
B					
A					

Battery Test pin



KYPD_VOLD_N_TP Factory test point



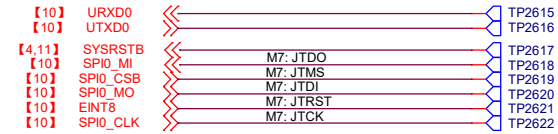
VBUS Test pin



Reserved for fast META mode detection



Debug test point



RAMDUMP debug key



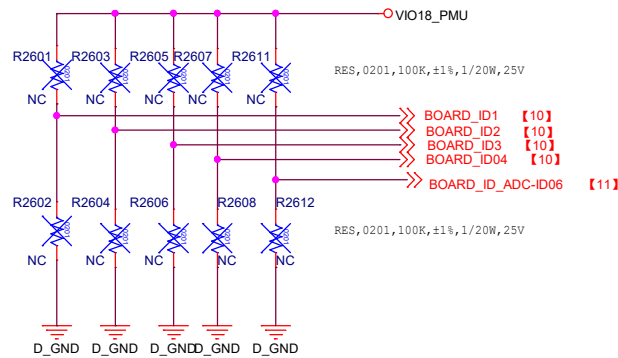
eMMC



IIC



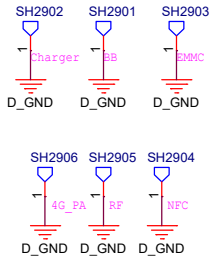
HW_ID



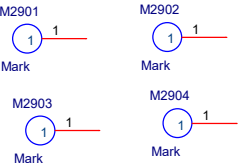


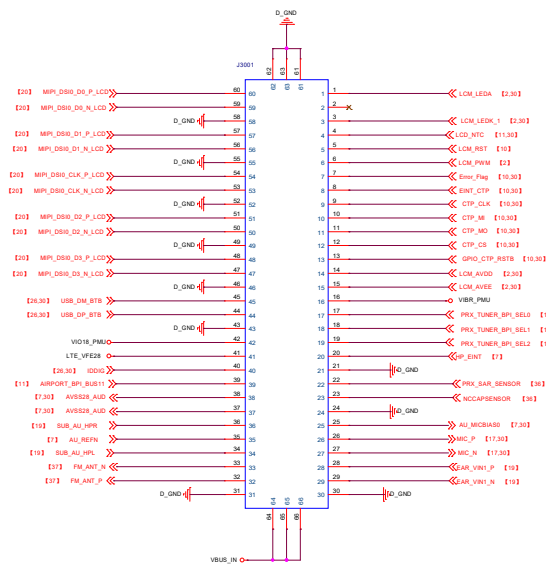
	5	4	3	2	1
D					
C					
B					
A					

Shielding frame

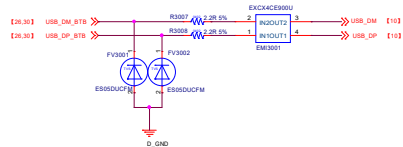
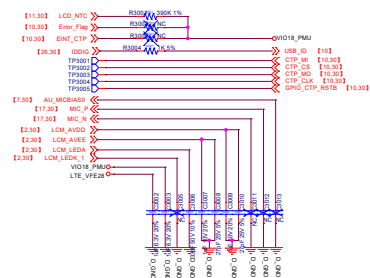


Marks point



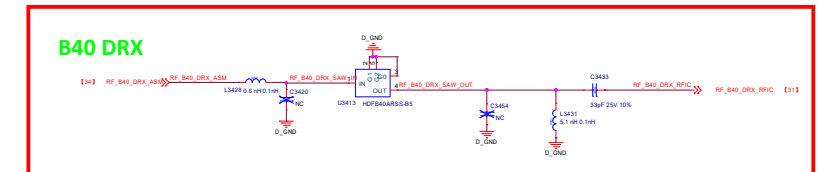
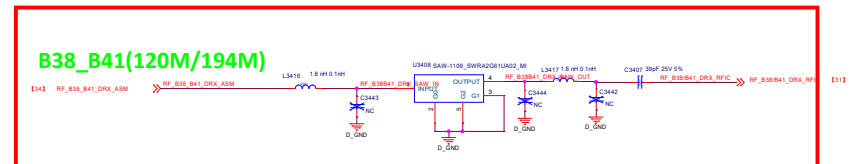
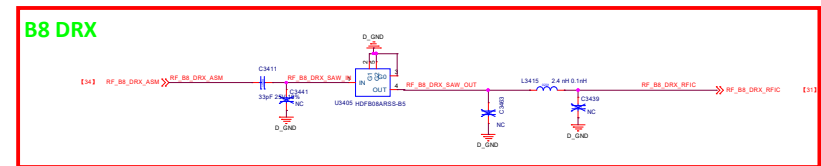
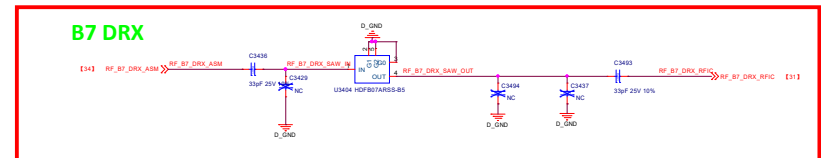
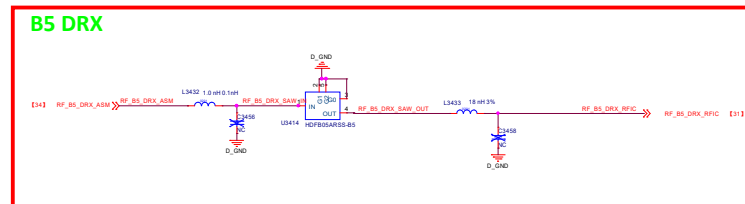
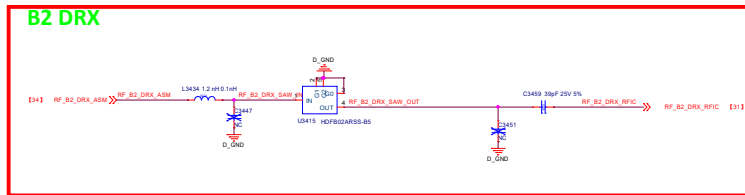
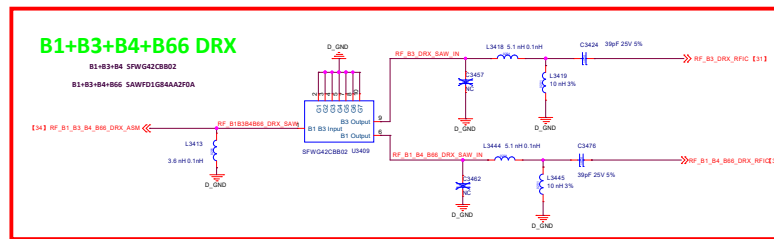
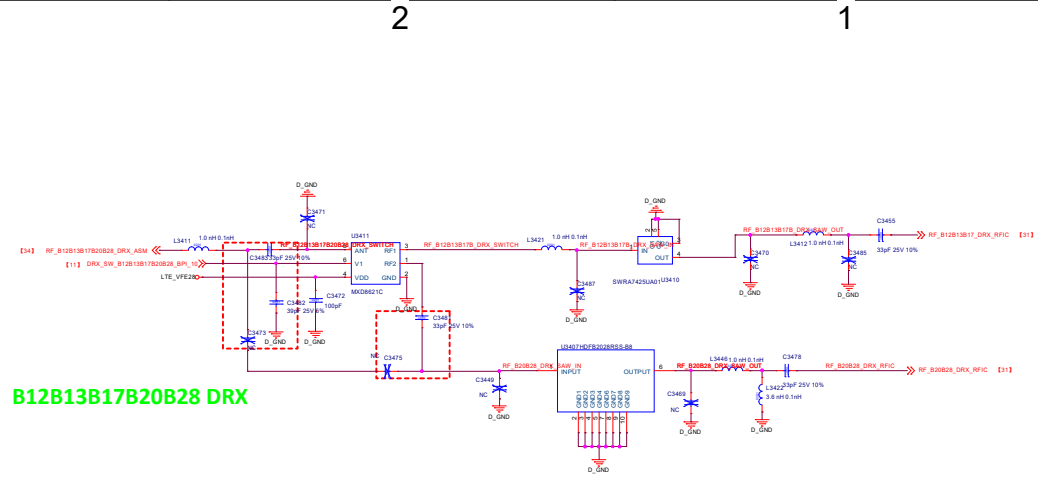
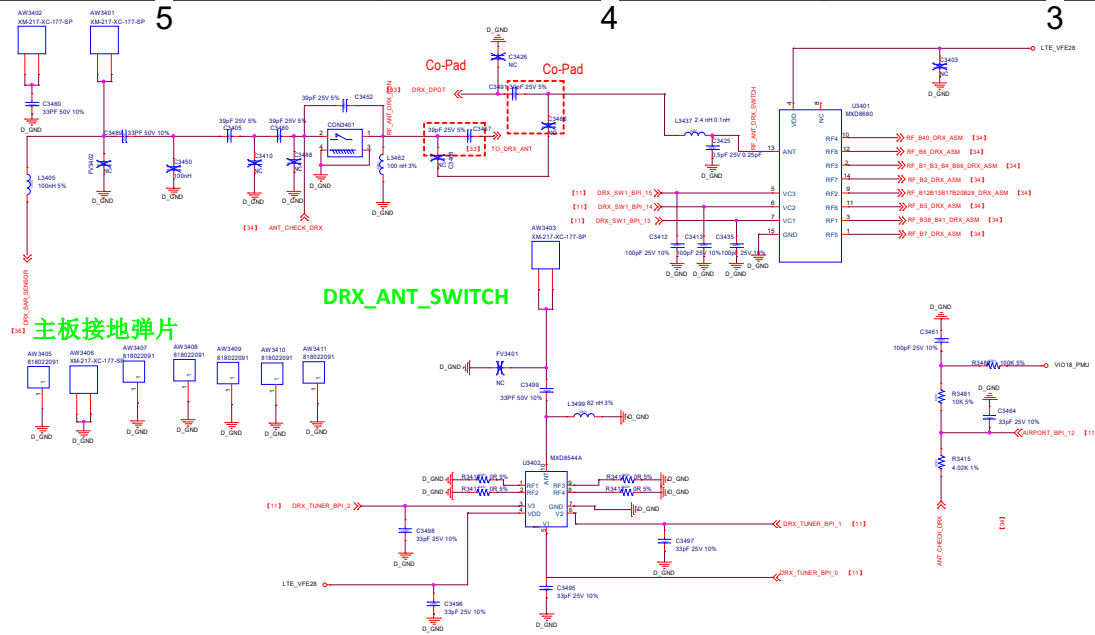


MAIN_SUB_CON

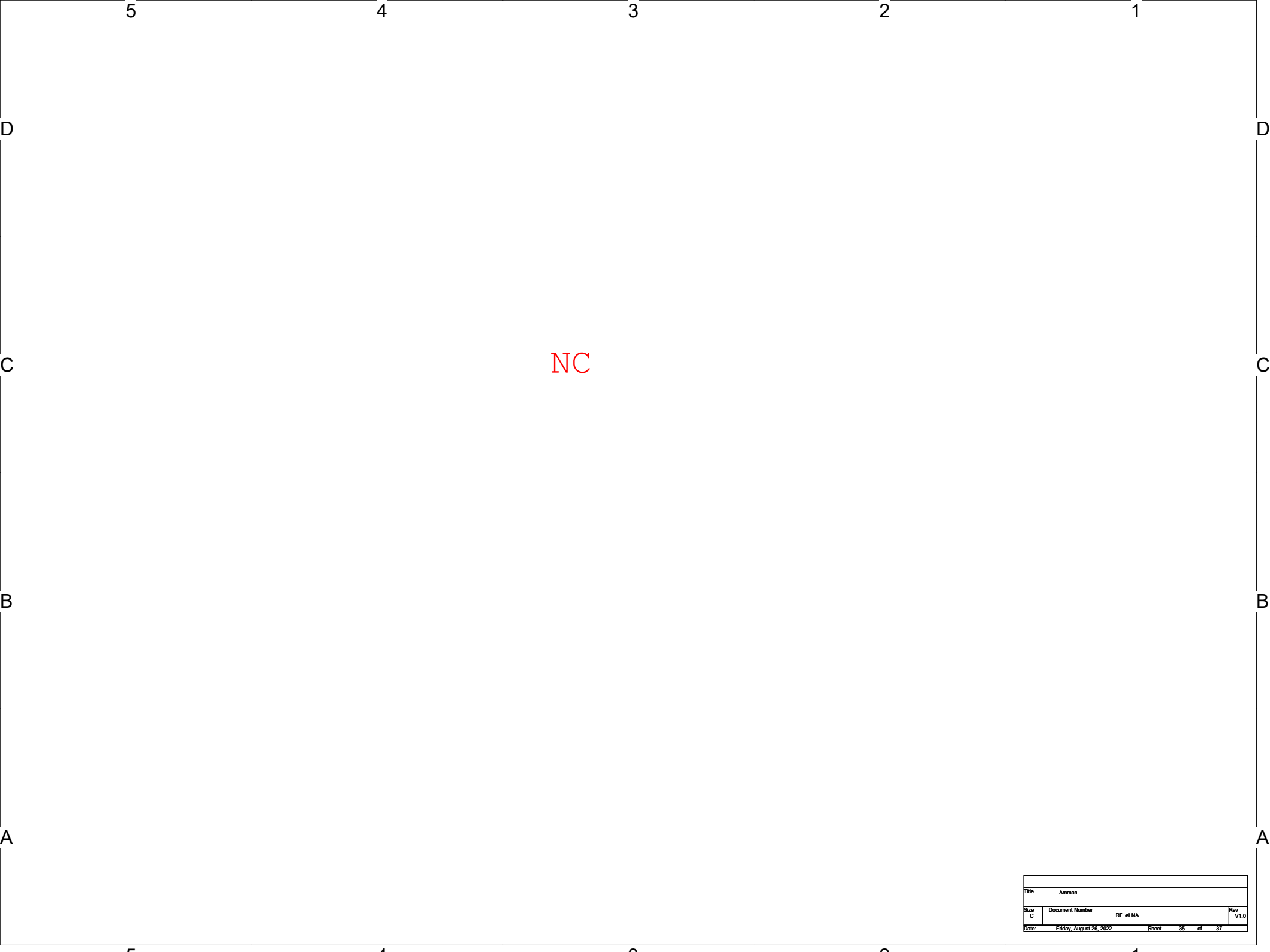




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Date:	Thursday, October 08, 2022	Sheet	32 of 37

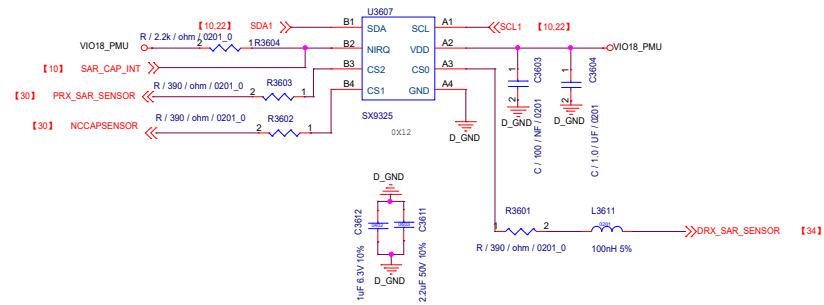


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File Amman			
Size C	Document Number RF_elNA		Rev V1.0
Date:	Friday, August 26, 2022		Sheet 35 of 37

Sar-Sensor



Title			
Amman			
File			
Amman			
Doc	Document Number	Capacitor	Rev
C	Document Number	Capacitor	Rev
Date	Friday, August 26, 2022	Sheet 36 of 37	1.0
Date: Friday, August 26, 2022 Sheet 36 of 37			

